

Laser Tag!

Right since the days of the first video games, directed energy weapons have captivated our imagination like no other form of weaponry out there. What fun would a game be if you had to shoot planet-destroying aliens with a .32 Beretta?

Almost all first-person and third-person sci-fi games use directed energy weapons like lasers as standard fare. There are pro-gamers in games like the *Quake* series (OK, this is our last mention of *Quake*!), the *Halo* series, *Gears of War*, and *Command & Conquer* who specialise in the use of these weapons. In these games, a directed-energy weapon zaps energy in a particular direction instead of like a projectile, for the desired blood and gore effect.

The story so far in real life, however, is different. Almost two decades since its inception in video games, these “lasers” have remained in the realm of the *Star Wars*



Some wonderful moments with laser guns in *Halo*



Some wonderful moments with laser guns in Israel

movie category. The only worldwide program that comes close is the Tactical High Energy Laser (THEL), which was designed and built by the Israeli government and is now being funded by the US Army Space & Missile Defense Command. The THEL is a high-energy laser weapons system that uses a heady mix of laser beam generation technologies and beam-

focusing technologies to merge with existing radar sensors and communication networks to provide a new defence theatre to ground-level, immobile assets. It can fire up to 60 shots without reloading, and if it meets design specification goals, the probability of kill of an absolute 100 per cent at a range of 4 km is guaranteed!

There is also talk of a mobile version of the THEL, conveniently baptised the M-THEL: the Mobile Tactical High Energy Laser, obviously, which can be mounted on an artillery vehicle and moved to forward posts if required.

Can't Touch This

Would you like to go into battle without any armour? No? Surprising—we feel the same way! However, the force-field type protection systems are going to do just that in the near future.

The force-field / force-shield concept has been floating around in academia for ages. It has been showcased in the *Star Trek* series, and more recently, *Halo 3* featured this protective system wherein a soldier could shield himself / herself (who says girls can't be soldiers?) from attack by forming a protective “bubble shield,” as they call it, around them.

Taking a cue from this, the RAFAEL Armament Development Authority of Israel, together with Israel Aircraft Industries' Elta Group, have developed the TROPHY Active Protection System (APS)—essentially a shield system for the armoured vehicles that protects

against incoming attacks. TROPHY intercepts and destroys incoming missiles and rockets with a blast similar to a shotgun-like explosion just before they hit the asset. Specifics of the materials and mechanics used in this explosive interceptor device are not available, but TROPHY is designed to form a “beam” of fragments that will intercept any incoming missile threat that gives off a lot of heat, like rockets, at a range of 15 to 35 metres from the targeted platform.

Since this system was designed to work in an urban environment, the direction, formation, and energy of the fragments used are made to ensure low collateral damage, and the lowest possible risk to nearby troops. The TROPHY system can simultaneously engage in multiple threats arriving from different directions, and is effective on stationary or moving platforms. The development program of TROPHY proposes an updated countermeasure unit to be available in the future, to protect against threat perceptions that only rely on kinetic energy to inflict damage against a traditional heat-based propulsion system.

As of April 2007, the TROPHY-APS has been deployed on tanks such as the Israeli Merkava Mk4. The developers claim that a system like TROPHY would reduce or eliminate the need for heavy armoured protection for combat vehicles.



The Bubble Shield in action in *Halo 3*



The TROPHY-APS System incorporated into a battle tank